

Causal vs passenger DNA methylation in a population-genetic model

Summary of the simulation phase (E0–E4)

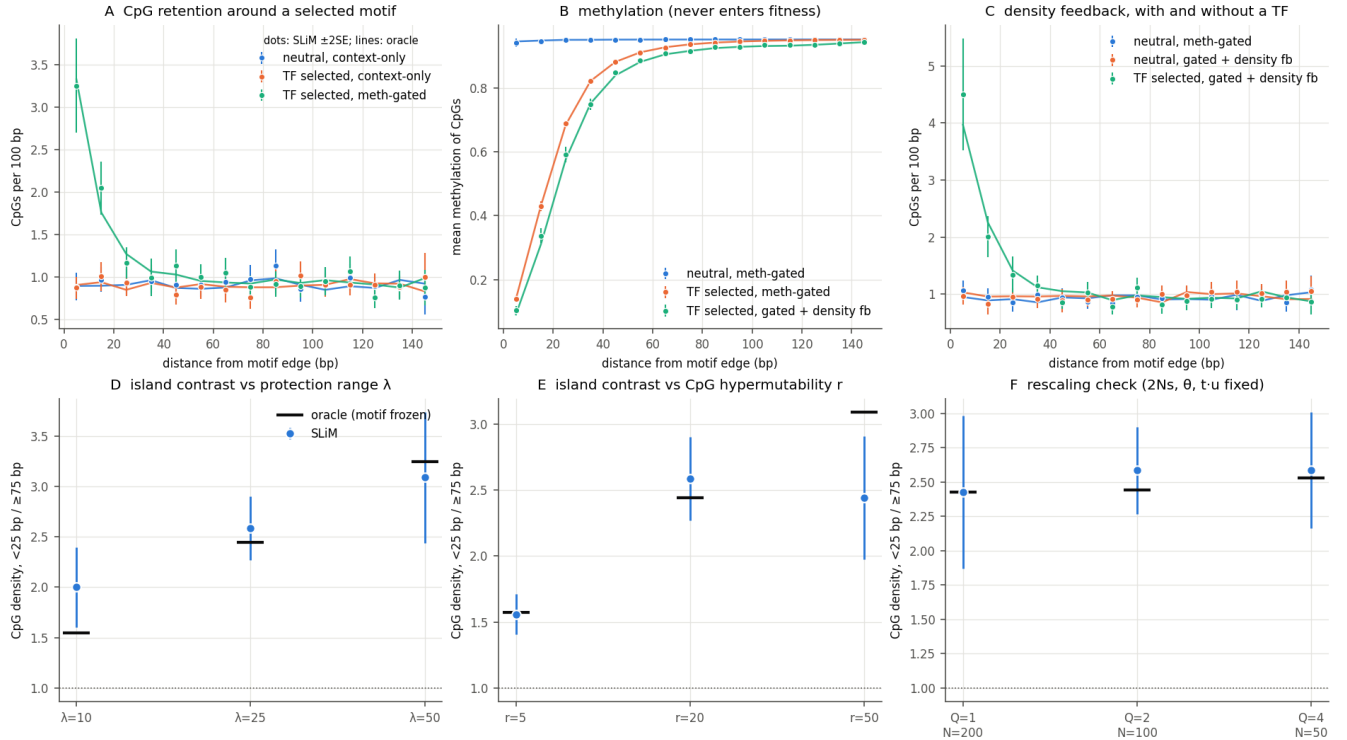
mcg_popgen working notes

2026-10-01

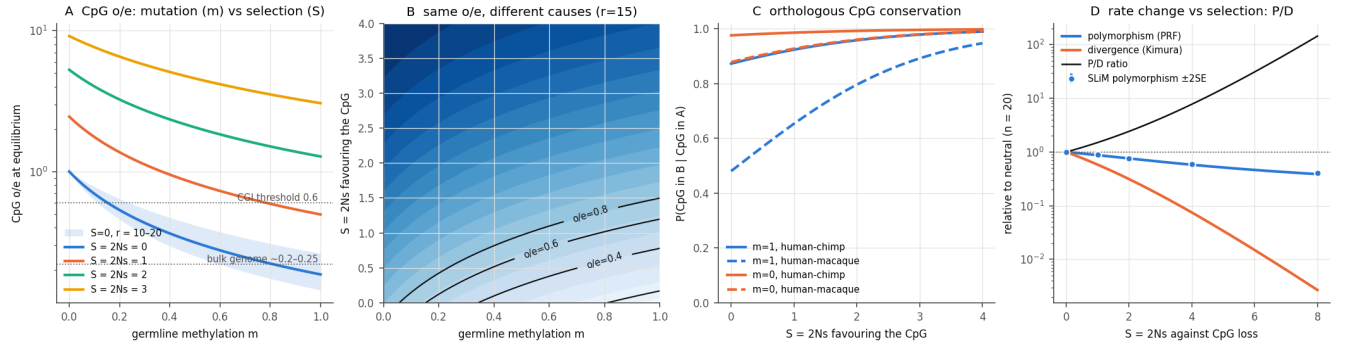
Question. Which population-genetic signatures distinguish *causal* methylation (fitness reads methylation) from *passenger* methylation (methylation is set by TF binding; fitness reads binding)? At what selection strength, and which survive realistic θ and meQTL detection?

Model. A Wright–Fisher population (SLiM 5.2) carries a 300-bp locus with one TF motif. Occupancy O and per-CpG methylation m are deterministic functions of each haplotype’s own sequence, and a bound TF protects nearby CpGs. Methylated CpGs mutate up to $r = 20\times$ faster. Fitness is stabilising, at an intermediate optimum, on O (passenger), on the methylation load $R = \sum m$ (causal), on both, or on $O(1 - m_{\text{motif CpG}})$ (methylation-sensitive TF). Results are reported as $2Ns$ and θ ; the headline runs use $\theta = 0.001$. Full specification: [pdf/theory.pdf](#). The simulator was validated against analytic results and an exact oracle (E0, 18/18 checks), with Python–SLiM parity $\leq 5 \times 10^{-7}$ and rescaling invariance.

1. Passive retention and the mutation–selection ridge

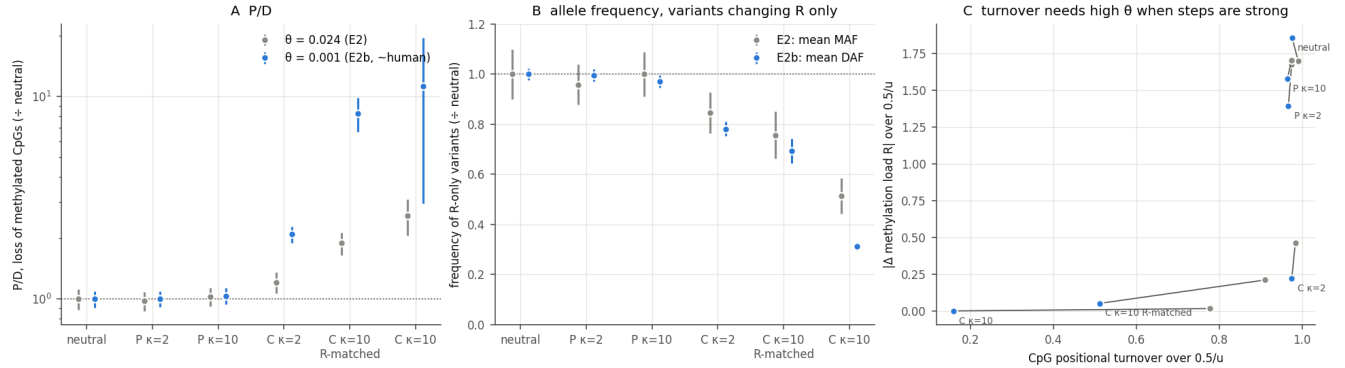


F1 (E1). With no selection on any CpG, a selected TF leaves a $2.6\times$ CpG enrichment within 25 bp, but only when mutation is methylation-gated (panel A, green vs orange). The footprint scales with protection width λ and with r (D, E), and rescaling holds (F).

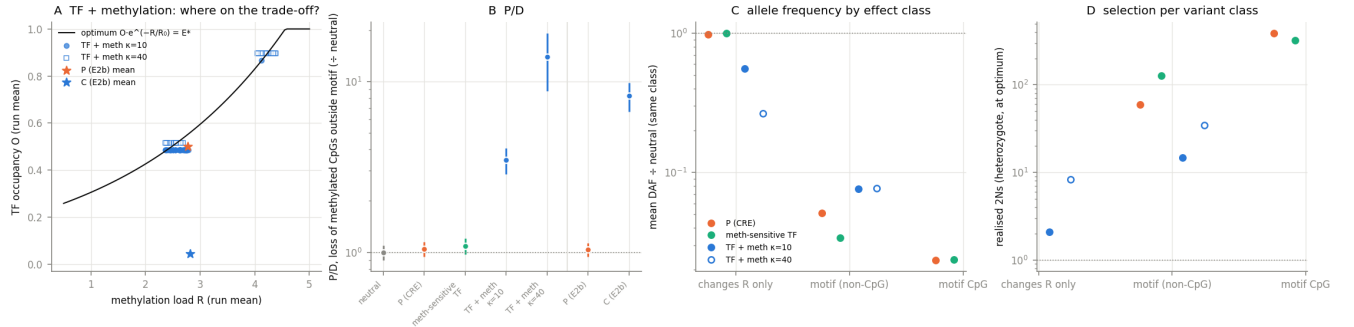


F2–F3 (P0). CpG o/e cannot tell germline hypomethylation from weak selection on CpGs (A, B): o/e 0.6 needs either $m \approx 0.15$ with no selection, or $S \approx 1.2$ on methylated CpGs. Cross-species conservation has the same ambiguity (C). The polymorphism/divergence ratio of CpG-loss alleles breaks the tie (D): $P/D = 1$ for a change in mutation rate, and rises with selection.

2. Causal vs passenger at realistic θ

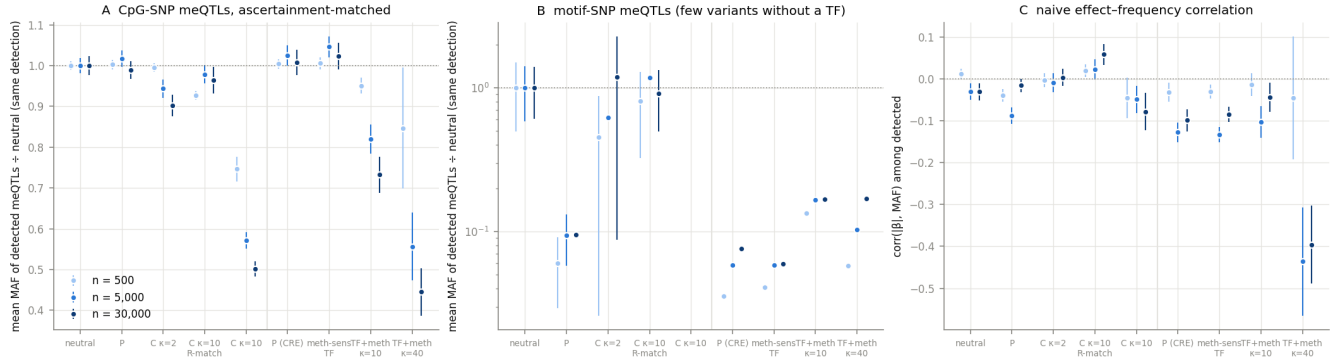


F3–F5, F8 (E2, E2b). Passenger methylation leaves no trace in CpG-only variants ($P/D \approx 1$, frequencies neutral). Causal methylation raises P/D to 2–11 and depletes those variants by 22–69%; the signals are stronger at human-like θ (A, B). Methylation conserved through CpG positional turnover requires weak per-CpG selection; strong steps freeze positions at realistic θ (C).



F5, F6, F9, F10 (E3). When TF and methylation are both causal, the locus shows both signatures: motif conserved and CpG-only variants depleted. Lineages commit to alternative (motif strength, methylation load) optima with the same expression (A). A methylation-sensitive TF is indistinguishable from its passenger control, because the bound TF self-protects its CpG.

3. Does it survive meQTL detection?



F4–F7 (E4). Compare detected CpG-SNP meQTLs with a neutral model put through the same detection (A). The ratio is 1.00 ± 0.05 under every passenger model, and $0.45\text{--}0.90$ at $n = 30,000$ under causal models with per-CpG $2Ns \gtrsim 2$. Motif-SNP depletion appears whenever a TF is selected (B). The naive effect–frequency correlation is as negative under passenger (detection bias) as under causal (C).

4. Predictions for real data

- R1. CpG excess at regulatory elements tracks *germline* hypomethylation; soma-only elements show none unless CpGs are selected or gBGC acts.
- R2. Stratified by germline methylation, CpG-loss $P/D = 1$ where only the mutation rate differs and > 1 where CpGs are selected. Non-CpG W $\leftrightarrow$ S sites control for gBGC.
- R3. CpG-SNP meQTLs outside motifs: ascertainment-matched MAF ratio = 1 under passenger, < 1 (falling with n) under causal. Include low-frequency variants.
- R4. Pooled meQTL effect–MAF correlations without a matched null are not evidence of selection.
- R5. Weak per-CpG constraint goes with CpG turnover at conserved methylation; strong constraint conserves CpG positions.
- R6. Matched expression is compatible with divergent motif strength and methylation, so methylation divergence is not expression divergence.
- R7. Sign tests on methylation divergence need a mutation-aware null that depends on each gene set’s CpG content and germline methylation (not yet simulated).

Limits. The biology is a toy: one locus, a hand-set map, cis only, no methylation–binding feedback. At $N = 100$ the minimum MAF is 0.005, so power in E4 is probably underestimated. The detection model is analytic, with no LD.

First cluster jobs: a check at real $N = 10^4$; real-data analysis B, or fitting the map to HPRC2 haplotype-resolved methylation (md/REAL_DATA.md, md/ENDPOINT.md).